

PMF, PDF, and CDF

General Lecture Notes on Representations of Probability Distributions

These notes are a conceptual and technical lecture on probability distributions and their representations. They are not solutions to an exercise list. The goal is to understand the general structures behind probability mass functions, probability density functions, cumulative distribution functions, supports, parameters, graphs, and probability calculations.

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1 Why Probability Distributions Need Representations

In probability theory we often start from a random experiment, but we do not always want to describe every elementary outcome directly. Instead, we usually define a random variable and study the probability law of that variable.

For example, in an experiment involving many individual components, the elementary outcome may contain detailed information about each component. However, the quantity of interest may be only the number of defective components. The random variable compresses the detailed outcome into a numerical value.

Random variable

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A real-valued random variable is a function

$$X : \Omega \rightarrow \mathbb{R}.$$

It assigns a real number to each elementary outcome. The distribution of X describes how probability is transferred from the original sample space Ω to values on the real line.

The probability distribution of a random variable can be represented in several related ways:

- by a **probability mass function** (PMF) for discrete random variables,
- by a **probability density function** (PDF) for continuous random variables,
- by a **cumulative distribution function** (CDF) for both discrete and continuous random variables,
- by tables, formulas, graphs, simulations, or numerical algorithms.

The central idea of this lecture is that these representations are different descriptions of the same object: the probability law of a random variable.

2 Experiment, Sample Space, Random Variable, and Support

Before working with a distribution, one should separate four levels of description.

1. The **random experiment**: what is being done or observed.
2. The **sample space** Ω : the set of elementary outcomes.
3. The **random variable** X : a numerical function defined on Ω .
4. The **support** of X : the set of values that X can take with positive probability or positive density.

Support

For a discrete random variable, the support is usually

$$\text{supp}(X) = \{x \in \mathbb{R} : \mathbb{P}(X = x) > 0\}.$$

For a continuous random variable with density f_X , the support is often described informally as the region where $f_X(x) > 0$, possibly including boundary points.

Do not confuse Ω and the support

The sample space Ω consists of elementary outcomes of the original experiment. The support consists of possible values of the random variable. These are not the same object.

Independent example: quality labels and a count variable

Suppose four produced items are classified as acceptable (A) or defective (D). An elementary outcome is a sequence such as

$$(A, D, A, A) \in \Omega.$$

The sample space contains all length-four sequences of A and D . Now define

$$X = \text{number of defective items.}$$

Then $X(A, D, A, A) = 1$. The support is

$$\{0, 1, 2, 3, 4\}.$$

Thus Ω contains sequences, while the support contains possible counts.

3 Discrete Random Variables and Probability Mass Functions

A discrete random variable takes values in a finite or countably infinite set. Its probability distribution can be described by assigning probability to each possible value.

Probability mass function

A probability mass function of a discrete random variable X is the function

$$p_X(x) = \mathbb{P}(X = x).$$

It assigns probability mass to individual values.

A valid PMF must satisfy two conditions:

$$p_X(x) \geq 0 \quad \text{for all } x,$$

and

$$\sum_{x \in \text{supp}(X)} p_X(x) = 1.$$

The PMF is the most direct representation of a discrete distribution. It immediately tells us the probability of exact values.

Independent example: number of failed logins

Let X denote the number of failed login attempts during a short session. Suppose the model is

x	0	1	2	3	4
$p_X(x)$	0.50	0.28	0.14	0.06	0.02

This is a valid PMF because all probabilities are nonnegative and

$$0.50 + 0.28 + 0.14 + 0.06 + 0.02 = 1.$$

For instance,

$$\mathbb{P}(X = 2) = 0.14, \quad \mathbb{P}(X \geq 3) = 0.06 + 0.02 = 0.08.$$

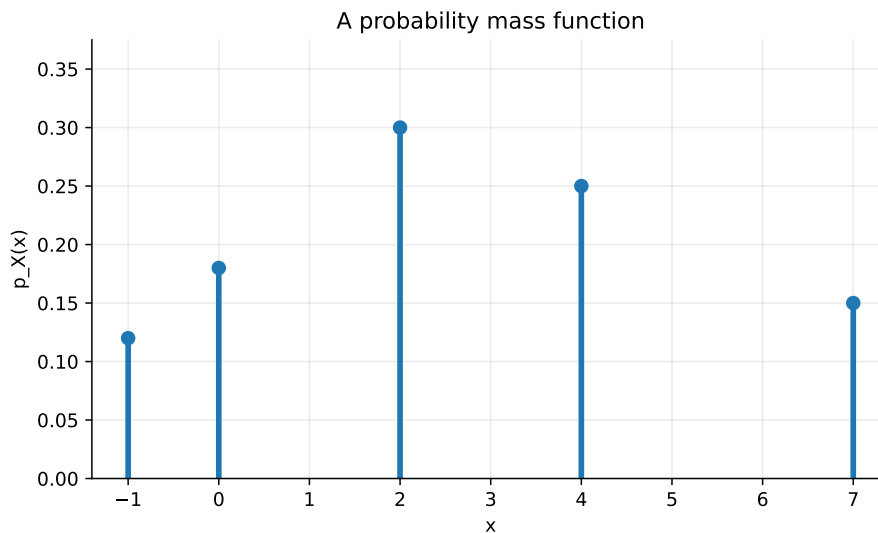


Figure 1: A PMF assigns probability mass to individual points.

3.1 Computing probabilities from a PMF

For a discrete random variable, the probability of an event involving X is computed by summing masses over the values that satisfy the event.

If $A \subseteq \mathbb{R}$, then

$$\mathbb{P}(X \in A) = \sum_{x \in A \cap \text{supp}(X)} p_X(x).$$

Common examples are:

$$\mathbb{P}(a < X \leq b) = \sum_{a < x \leq b} p_X(x),$$

$$\mathbb{P}(X \geq a) = \sum_{x \geq a} p_X(x),$$

$$\mathbb{P}(X \neq a) = 1 - p_X(a).$$

4 Cumulative Distribution Functions

The cumulative distribution function is the most universal representation of the distribution of a real-valued random variable.

Cumulative distribution function

The cumulative distribution function of a random variable X is

$$F_X(x) = \mathbb{P}(X \leq x), \quad x \in \mathbb{R}.$$

The CDF answers the question: *How much probability is accumulated up to the point x ?*

Every CDF has the following properties:

Basic properties of a CDF

A function $F : \mathbb{R} \rightarrow [0, 1]$ is a CDF only if it has the following properties:

1. F is nondecreasing.
2. $\lim_{x \rightarrow -\infty} F(x) = 0$.
3. $\lim_{x \rightarrow +\infty} F(x) = 1$.
4. F is right-continuous.

For a discrete distribution, the CDF is a step function. At each point where there is probability mass, the CDF jumps upward.

Jump size equals probability mass

For a discrete random variable,

$$\mathbb{P}(X = a) = F_X(a) - \lim_{x \rightarrow a^-} F_X(x).$$

That is, the probability at a is exactly the size of the jump of the CDF at a .

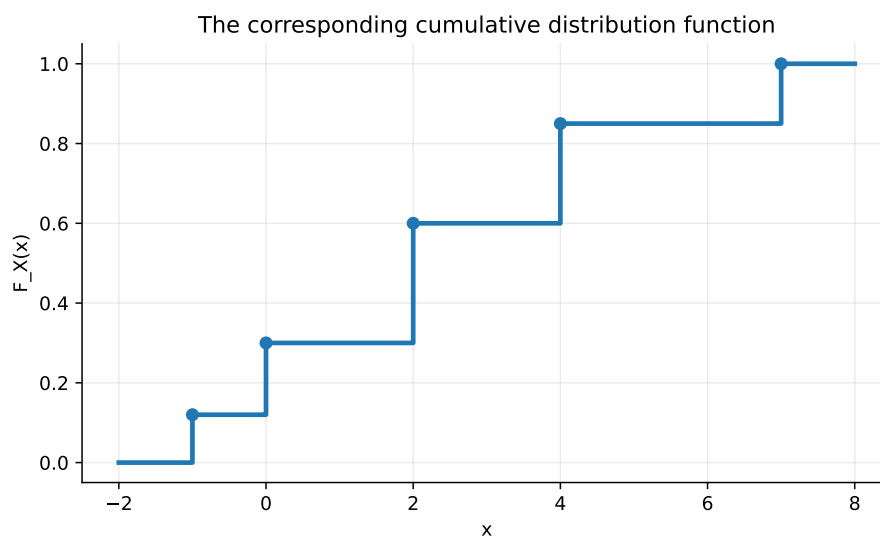


Figure 2: For a discrete distribution, the CDF is a step function.

4.1 Computing probabilities from a CDF

The CDF is especially useful for interval probabilities. For any random variable,

$$\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a).$$

For continuous random variables, endpoint distinctions usually do not matter, but for discrete random variables they do. In the discrete case,

$$\mathbb{P}(a \leq X \leq b) = F_X(b) - \mathbb{P}(X < a).$$

If a is a support point, then

$$\mathbb{P}(a \leq X \leq b) = F_X(b) - F_X(a^-),$$

where

$$F_X(a^-) = \lim_{x \rightarrow a^-} F_X(x).$$

Strict and non-strict inequalities

In discrete probability, $X < a$ and $X \leq a$ may give different probabilities. In continuous probability, they usually give the same probability because individual points have probability zero. This difference is one of the most common sources of mistakes.

5 Continuous Random Variables and Probability Density Functions

A continuous random variable does not assign positive probability to individual values. Instead, probabilities are computed as areas under a density curve.

Probability density function

A probability density function of a continuous random variable X is a function $f_X : \mathbb{R} \rightarrow [0, \infty)$ such that

$$\int_{-\infty}^{\infty} f_X(x) \, dx = 1,$$

and probabilities are computed by

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) \, dx.$$

A density is not itself a probability. A density value may be larger than 1. What must equal 1 is the total area under the curve.

Relationship between PDF and CDF

If X has density f_X , then

$$F_X(x) = \int_{-\infty}^x f_X(t) \, dt.$$

At points where F_X is differentiable,

$$f_X(x) = F'_X(x).$$

For continuous distributions,

$$\mathbb{P}(X = a) = 0.$$

Therefore,

$$\mathbb{P}(a < X < b) = \mathbb{P}(a \leq X \leq b) = \mathbb{P}(a < X \leq b) = F_X(b) - F_X(a).$$

Independent example: measurement error

Suppose X denotes the measurement error of a sensor in millimeters. If the error is modeled continuously, the probability that X is exactly $0.100000\dots$ mm is zero. But the probability that the error lies between -0.1 and 0.1 mm may be positive:

$$\mathbb{P}(-0.1 \leq X \leq 0.1) = \int_{-0.1}^{0.1} f_X(x) dx.$$

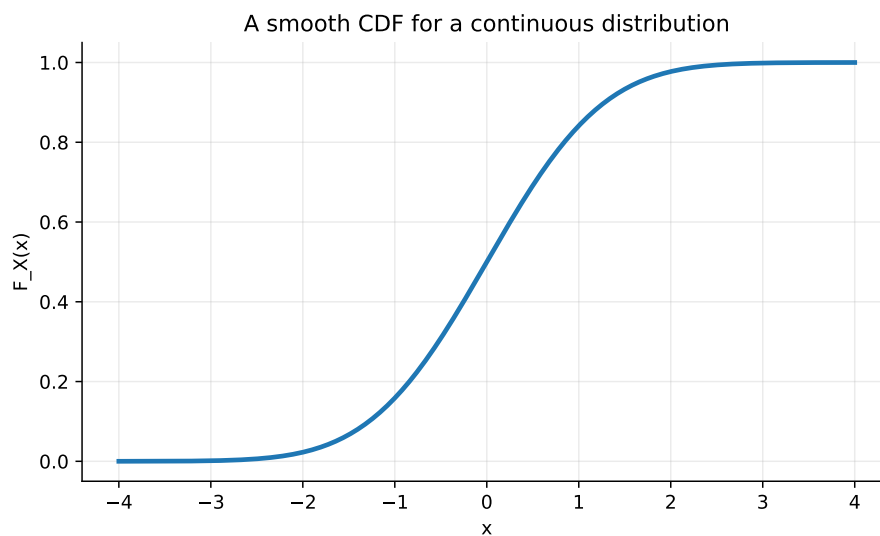


Figure 3: A continuous distribution often has a smooth CDF.

6 PMF, PDF, and CDF Compared

Representation	Main use	Typical object	How probabilities are computed
PMF $p_X(x)$	Exact values	Discrete variables	Sum probability masses
PDF $f_X(x)$	Density and shape	Continuous variables	Integrate over intervals
CDF $F_X(x)$	Accumulated probability	Discrete or continuous variables	Take differences $F(b) - F(a)$

The PMF and PDF are local descriptions. They show where probability mass or density is concentrated. The CDF is cumulative. It shows how much probability has accumulated up to a given point.

A useful reading order

When studying a distribution, ask:

1. What random experiment is being modeled?
2. What is the random variable?
3. What is the support?
4. Is the variable discrete or continuous?
5. Which representation is given: PMF, PDF, or CDF?
6. Which representation is best for the required probability?
7. How do the parameters change the graph?

7 Discrete Distribution Families

This section summarizes important discrete distributions from the point of view of PMF, CDF, support, assumptions, and graph shape.

7.1 Bernoulli distribution

The Bernoulli distribution models one trial with two possible outcomes: success and failure.

$$X \sim \text{Bernoulli}(p), \quad X \in \{0, 1\}.$$

The PMF is

$$\mathbb{P}(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}.$$

Here p is the probability of success. The expectation and variance are

$$\mathbb{E}[X] = p, \quad \text{Var}(X) = p(1 - p).$$

Independent example: one system check

Let $X = 1$ if an automated diagnostic test detects a warning and $X = 0$ otherwise. If the warning probability is $p = 0.08$, then $X \sim \text{Bernoulli}(0.08)$.

7.2 Binomial distribution

The binomial distribution models the number of successes in a fixed number of independent Bernoulli trials with constant success probability.

Binomial distribution

If $X \sim \text{Bin}(n, p)$, then

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n.$$

The support is

$$\{0, 1, \dots, n\}.$$

The parameters are:

- n : number of trials,
- p : probability of success in each trial.

The expectation and variance are

$$\mathbb{E}[X] = np, \quad \text{Var}(X) = np(1-p).$$

When p increases, the mass shifts to the right. When n increases, the support becomes wider and the distribution usually becomes more spread out in absolute terms, although relative fluctuations may become smaller.

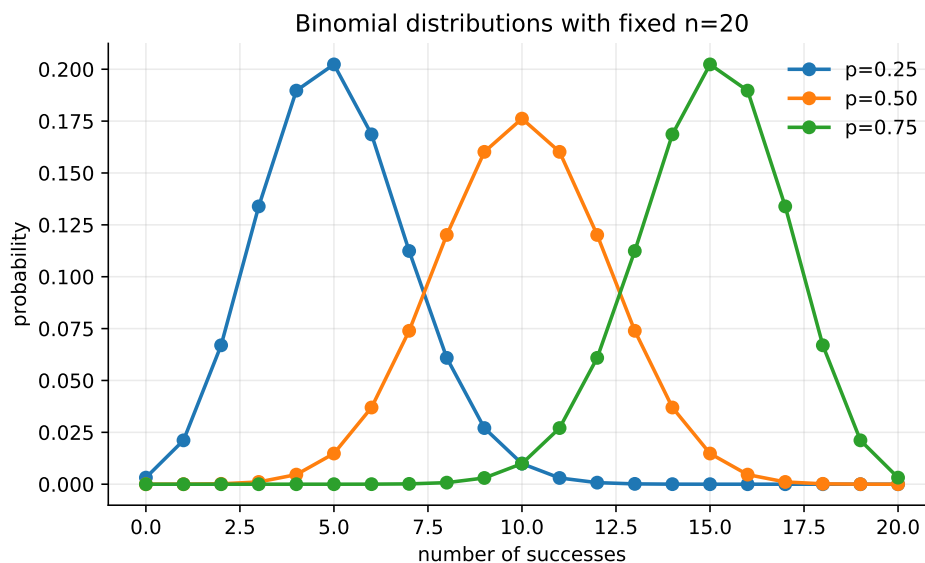


Figure 4: Changing p shifts a binomial distribution.

Binomial assumptions

The binomial model requires a fixed number of trials, two outcomes per trial, independence, and constant success probability. If sampling is without replacement from a small finite population, the hypergeometric model is usually more appropriate.

7.3 Geometric distribution

The geometric distribution models the waiting time until the first success in independent Bernoulli trials.

Geometric distribution

Using the convention that X is the trial number of the first success,

$$X \sim \text{Geom}(p),$$

with support

$$\{1, 2, 3, \dots\},$$

and PMF

$$\mathbb{P}(X = k) = (1 - p)^{k-1}p, \quad k = 1, 2, 3, \dots$$

The CDF is

$$F_X(k) = \mathbb{P}(X \leq k) = 1 - (1 - p)^k, \quad k = 1, 2, 3, \dots$$

The tail probability has a simple interpretation:

$$\mathbb{P}(X > k) = (1 - p)^k.$$

It means that no success occurred in the first k trials.

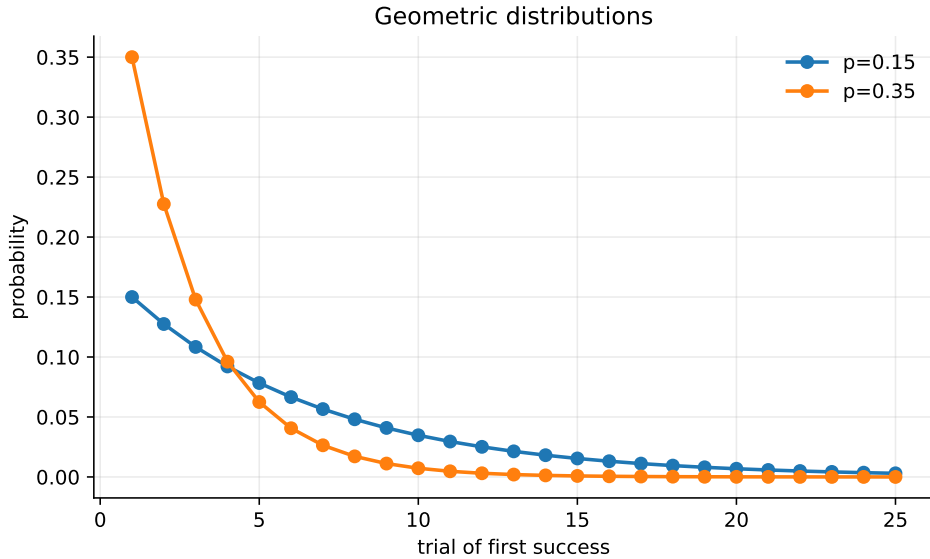


Figure 5: For larger p , the first success tends to occur earlier.

7.4 Negative binomial distribution

The negative binomial distribution generalizes the geometric distribution. Instead of waiting for the first success, we wait for the r -th success.

Negative binomial distribution

Let X be the trial number on which the r -th success occurs. Then

$$\mathbb{P}(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k = r, r+1, r+2, \dots$$

The binomial coefficient counts the positions of the first $r-1$ successes among the first $k-1$ trials. The final trial must be a success.

Independent example: repeated password recovery attempts

A system may record attempts until a user completes the third successful verification. If each attempt succeeds independently with probability p , then the number of attempts needed can be modeled by a negative binomial distribution.

7.5 Poisson distribution

The Poisson distribution models the number of events in a fixed interval of time, space, length, area, or volume, under assumptions of random independent occurrence with a constant average rate.

Poisson distribution

If $X \sim \text{Pois}(\lambda)$, then

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

The support is

$$\{0, 1, 2, \dots\}.$$

The parameter $\lambda > 0$ is the expected number of events in the interval. For a Poisson distribution,

$$\mathbb{E}[X] = \lambda, \quad \text{Var}(X) = \lambda.$$

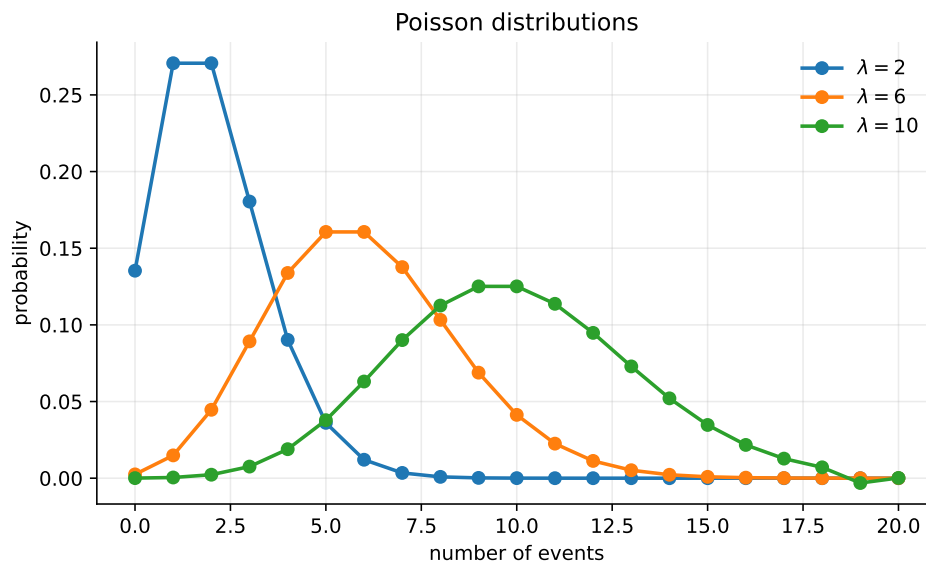


Figure 6: As λ increases, the Poisson distribution shifts right and becomes more spread out.

Independent example: messages arriving at a server

If a monitoring system receives on average 4 warning messages per minute and the assumptions of the Poisson model are reasonable, then the number of warnings in one minute can be modeled as $X \sim \text{Pois}(4)$.

7.6 Hypergeometric distribution

The hypergeometric distribution models sampling without replacement from a finite population containing two types of objects.

Hypergeometric distribution

Suppose a population contains N objects, of which K are distinguished. A sample of size n is drawn without replacement. If X is the number of distinguished objects in the sample, then

$$\mathbb{P}(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}.$$

The possible values of k must satisfy

$$\max(0, n - (N - K)) \leq k \leq \min(n, K).$$

The hypergeometric model differs from the binomial model because trials are not independent: after one object is drawn, the composition of the remaining population changes.

Binomial or hypergeometric?

Use a binomial model when trials are independent with constant success probability. Use a hypergeometric model when sampling is without replacement from a finite population and the population composition matters.

8 Continuous Distribution Families

Continuous distributions are described by PDFs and CDFs. Their graphs are especially important because probabilities are areas.

8.1 Uniform distribution

The continuous uniform distribution on $[a, b]$ assigns equal density to every point in the interval.

$$X \sim U(a, b), \quad f_X(x) = \frac{1}{b-a}, \quad a \leq x \leq b.$$

The CDF is

$$F_X(x) = \begin{cases} 0, & x < a, \\ \frac{x-a}{b-a}, & a \leq x \leq b, \\ 1, & x > b. \end{cases}$$

Independent example: randomized delay

A system intentionally waits for a random delay uniformly chosen between 2 and 5 seconds. The delay can be modeled as $U(2, 5)$.

8.2 Beta distribution

The beta distribution is a flexible family on the interval $[0, 1]$. It is often used to model proportions, probabilities, and fractions.

Beta distribution

If $X \sim \text{Beta}(\alpha, \beta)$, with $\alpha > 0$ and $\beta > 0$, then

$$f_X(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 < x < 1,$$

where

$$B(\alpha, \beta) = \int_0^1 t^{\alpha-1} (1-t)^{\beta-1} dt$$

is the beta function.

The parameters control the shape. Depending on α and β , the density may be symmetric, skewed, U-shaped, or concentrated around the center.

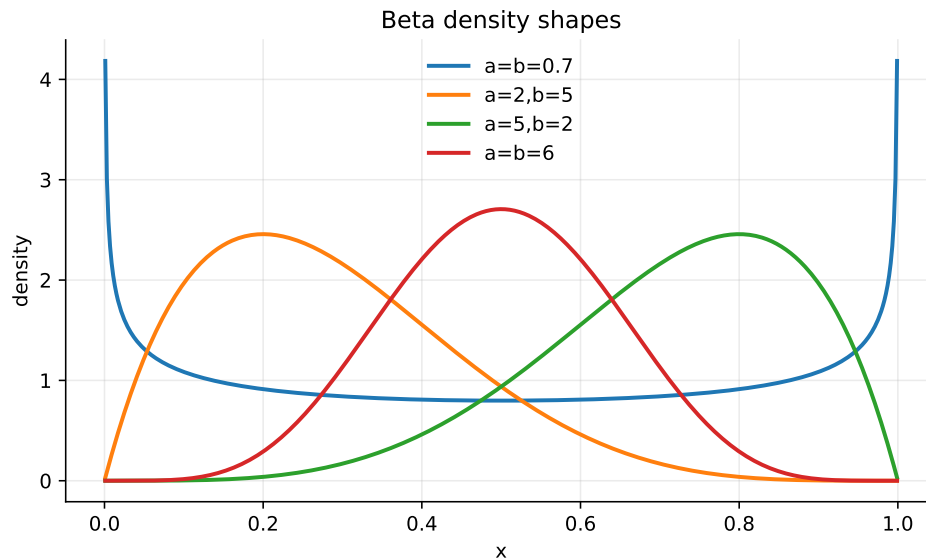


Figure 7: The beta family is flexible on the interval $[0, 1]$.

8.3 Gamma distribution and chi-square distribution

The gamma distribution is used for positive continuous quantities such as waiting times, lifetimes, and accumulated positive variation.

Gamma distribution

With shape $\alpha > 0$ and scale $\theta > 0$,

$$X \sim \text{Gamma}(\alpha, \theta)$$

has density

$$f_X(x) = \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-x/\theta}, \quad x > 0.$$

The expectation and variance are

$$\mathbb{E}[X] = \alpha\theta, \quad \text{Var}(X) = \alpha\theta^2.$$

The chi-square distribution is a special case:

$$\chi_\nu^2 = \text{Gamma}\left(\frac{\nu}{2}, 2\right).$$

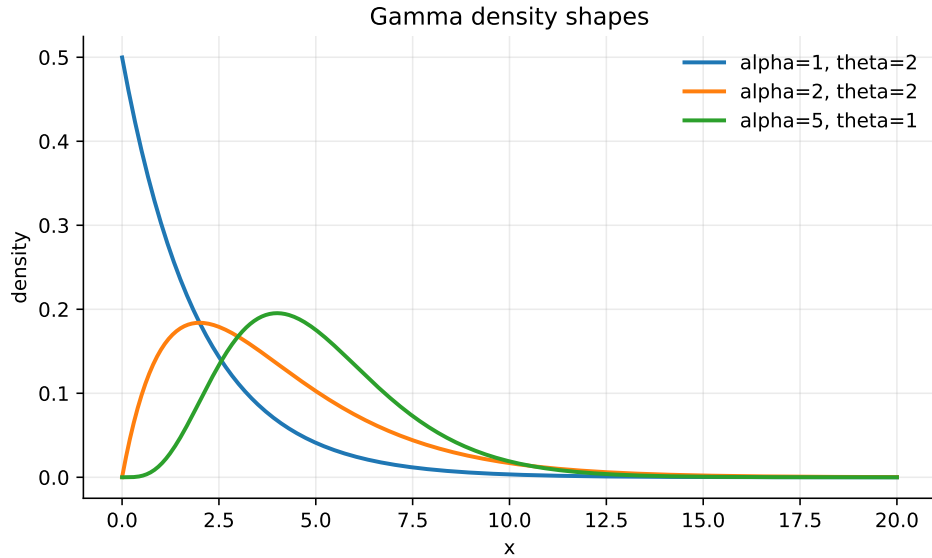


Figure 8: Gamma densities are supported on positive values and can model waiting-time quantities.

8.4 Normal distribution

The normal distribution is one of the most important continuous models. It is used for measurement errors, aggregate fluctuations, standardized scores, and many approximate models.

Normal distribution

If $X \sim N(\mu, \sigma^2)$, then

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

The parameter μ controls location, and σ controls spread. The variance is σ^2 .

To compute probabilities for a normal distribution, one often standardizes:

$$Z = \frac{X - \mu}{\sigma}.$$

Then $Z \sim N(0, 1)$, and

$$\mathbb{P}(a \leq X \leq b) = \mathbb{P}\left(\frac{a - \mu}{\sigma} \leq Z \leq \frac{b - \mu}{\sigma}\right).$$

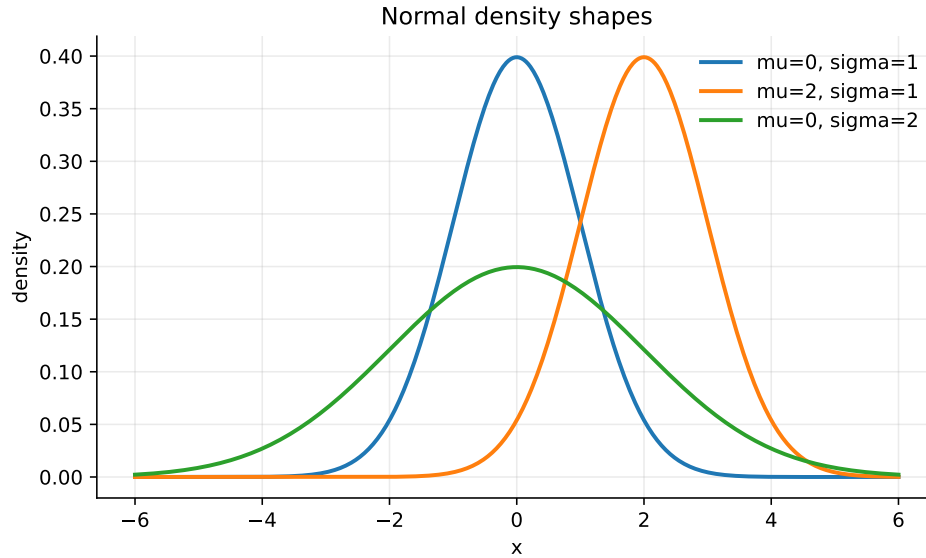


Figure 9: Changing μ shifts a normal density; changing σ changes its spread.

9 Moving Between Representations

A major skill in probability is moving between PMF/PDF and CDF representations.

9.1 From PMF to CDF

For a discrete random variable,

$$F_X(x) = \sum_{t \leq x} p_X(t).$$

This means that each value of the CDF is obtained by accumulating all masses at or to the left of x .

9.2 From CDF to PMF

For a discrete random variable, the PMF can be recovered from jumps:

$$p_X(a) = F_X(a) - F_X(a^-).$$

If the CDF is given in a table, then successive differences often reveal the mass at each support point.

9.3 From PDF to CDF

For a continuous random variable,

$$F_X(x) = \int_{-\infty}^x f_X(t) dt.$$

This is a cumulative area under the density curve.

9.4 From CDF to PDF

If the CDF is differentiable at x , then

$$f_X(x) = F'_X(x).$$

A CDF that is steep in some region corresponds to high density in that region.

A density is not a probability

It is wrong to interpret $f_X(3)$ as $\mathbb{P}(X = 3)$. For a continuous random variable, $\mathbb{P}(X = 3) = 0$. The value $f_X(3)$ describes local density, not probability at a point.

10 Graphs and Parameter Interpretation

Graphs are not decoration. For distributions, graphs reveal structure.

10.1 PMF graphs

A PMF graph should emphasize individual support points. Good representations include vertical sticks, dots, or bar plots. The total height is not meaningful by itself; the sum of the masses is meaningful.

When comparing PMFs, use the same horizontal scale and vertical scale whenever possible.

10.2 PDF graphs

A PDF graph should be interpreted by area, not by height alone. A higher density over a very narrow region may contain less probability than a lower density over a wide region.

When comparing PDFs, look at:

- location,
- spread,
- symmetry or skewness,
- number of modes,
- tail behavior,
- support boundaries.

10.3 CDF graphs

A CDF graph always increases from 0 to 1. It reveals:

- where probability accumulates quickly,
- whether the distribution is discrete or continuous,
- medians and quantiles,

- tail probabilities.

Quantile

A q -quantile is a value x_q such that approximately

$$F_X(x_q) = q.$$

For example, a median is a 0.5-quantile.

11 Model Choice and Distribution Assumptions

A distribution is not just a formula. It encodes assumptions about the random experiment.

Distribution	What it models	Key assumptions
Bernoulli	one success/failure trial	two outcomes, fixed p
Binomial	number of successes in n trials	independent trials, constant p
Geometric	waiting time to first success	independent trials, constant p
Negative binomial	waiting time to r -th success	independent trials, constant p
Poisson	number of events in an interval	constant rate, independent occurrence structure
Hypergeometric	successes in sampling without replacement	finite population, changing composition
Uniform	equally likely values on an interval	constant density on support
Beta	proportions on $[0, 1]$	flexible shape on bounded interval
Gamma	positive waiting-time quantities	positive support, shape-scale behavior
Normal	symmetric measurement-like variation	real support, bell-shaped symmetry

Model selection questions

Before choosing a distribution, ask:

1. Is the random variable discrete or continuous?
2. What is the support?
3. Is it a count, a waiting time, a proportion, or a measurement?
4. Are trials independent?
5. Is the number of trials fixed or random?
6. Is sampling with or without replacement?
7. Are values bounded, positive, or unrestricted on the real line?

12 Exact Values, Intervals, and Tail Probabilities

Different representations are convenient for different types of probability questions.

12.1 Exact values

For a discrete random variable,

$$\mathbb{P}(X = k) = p_X(k).$$

For a continuous random variable,

$$\mathbb{P}(X = k) = 0.$$

12.2 Left-tail probabilities

The CDF gives left-tail probabilities immediately:

$$\mathbb{P}(X \leq a) = F_X(a).$$

12.3 Right-tail probabilities

Right-tail probabilities are computed by complements:

$$\mathbb{P}(X > a) = 1 - F_X(a).$$

For continuous variables,

$$\mathbb{P}(X \geq a) = 1 - F_X(a)$$

as well. For discrete variables, be careful with the mass at a .

12.4 Interval probabilities

For all real-valued random variables,

$$\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a).$$

For continuous variables, endpoint choices do not change the probability.

13 Computational and Visual Work

Modern probability work often involves computational tools. However, computation should support mathematical understanding, not replace it.

A good computational study of a distribution should include:

- the formula of the PMF or PDF,
- the support,
- parameter values,
- a PMF/PDF graph,
- a CDF graph,
- selected probability calculations,
- an interpretation of how the parameters affect the graph.

Common plotting mistakes

Avoid these mistakes:

- drawing a discrete PMF as if it were a continuous curve without explaining the approximation,
- interpreting PDF height as probability,
- forgetting that a CDF must be nondecreasing,
- using different scales when comparing distributions and then drawing false conclusions,
- ignoring support boundaries.

14 Short Conceptual Summary

- A random variable maps elementary outcomes to numbers.
- The support is the set of possible values of the random variable, not necessarily the original sample space.
- A PMF assigns probabilities to individual values of a discrete random variable.
- A PDF assigns density to a continuous random variable; probabilities are areas under the curve.
- A CDF gives accumulated probability: $F_X(x) = \mathbb{P}(X \leq x)$.
- In a discrete distribution, CDF jumps equal probability masses.
- In a continuous distribution, the PDF is the derivative of the CDF where the derivative exists.
- Exact point probabilities are meaningful for discrete variables but zero for continuous variables.
- Parameters are not just symbols; they control location, spread, skewness, tails, and support.
- Choosing a distribution means choosing assumptions about the random experiment.

15 Checklist for Studying Any Distribution

For any probability distribution, students should be able to answer the following questions.

1. What experiment or phenomenon can the distribution model?
2. What is the random variable?
3. Is the random variable discrete or continuous?
4. What is the support?
5. What is the PMF or PDF?
6. What is the CDF?
7. Which parameters appear, and what do they mean?
8. How do parameter changes affect the graph?
9. How are probabilities of exact values, intervals, and tails computed?

10. What are common applications and common misuses of the model?

Learning goals after this lecture

After studying these notes, a student should be able to look at a probability distribution and explain it structurally: where it comes from, what values it can take, how its probabilities are represented, how its graph should be read, and which probability calculations are natural for that representation. The main skill is not memorizing formulas separately, but seeing PMF, PDF, and CDF as connected descriptions of one probability law.